COURSE MATERIAL	
SUBJECT	**ENGINEERING PHYSICS(PH23ABS101)**
UNIT	**II**
COURSE	**B.TECH**
DEPARTMENT	**ECE**
SEMESTER	**I-II**
PREPARED BY (Faculty Name/s)	**P.GIRIJA** **Assistant Professor**
Version	**V-2**
PREPARED / REVISED DATE	**30-05-2023**

TABLE OF CONTENTS – UNIT IV

1. **Course Objectives**

The objectives of this course is to

Bridge the gap between the Physics in school at 10+2 level and UG level engineering courses by identifying the importance of the optical phenomenon like interference, diffraction etc, enlightening the periodic arrangement of atoms in crystalline solids and concepts of quantum mechanics, introduce novel concepts of dielectric and magnetic materials, physics of semiconductors.

2. **Prerequisites**

Students should have knowledge on

1. Wave Optics
2. Crystallography
3. Quantum Mechanics
4. Semiconductors

3. **Syllabus**

UNIT II

Crystallography- Space lattice, Basis, unit cell and lattice parameters – Crystal systems –Bravais Lattices – Crystal systems – Packing fraction – Coordination number – Packing fraction of SC,BCC & FCC – Nacl structure – Miller indices – Separation between successive (hkl) planes.

X-Ray Diffraction- Diffraction of X-rays by crystal planes - Bragg's law – Crystal structure determination by Laue and Powder methods.

4. **Course outcomes**

1. CO 1: Understand the intensity variation of light due to interference, diffraction and polarization.
2. CO 2: Apply the basic concepts of crystal structures and X-ray diffraction to study the behavior of materials for engineering applications.
3. CO 3: Summarize the fundamental properties of dielectric and magnetic materials for engineering applications.
4. CO 4: Analyze the properties of quantum particles to interpret the energy band formation and classification of solids
5. CO 5: Assess the current flow mechanism to understand the transport phenomenon of charge carriers in semiconductors.

5. Co-PO / PSO Mapping

APPLIED PHYSICS	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	3	3												
CO2	3	3												
CO3	3	3												
CO4	3	3												
CO5	3	3												

6. Lesson Plan

Lecture No.	Weeks	Topics to be covered	References
1	1	Introduction	T1, R1
2		Space lattice, Basis, unit cell and lattice parameters	T1, R1
3		Crystal systems	T1, R1
4		Bravais Lattices	T1, R1
5		Crystal systems	T1, R1
6	2	SCC	T1, R1
7		BCC	T1, R1
8		FCC	T1, R1
9		Nacl structure	T1, R1
10	3	Miller indices	T1, R1
11		Separation between successive (hkl) planes.	T1, R1
12		X-ray diffraction introduction	T1, R1
13		Diffraction of X-rays by crystal planes	T1, R1
14	4	Bragg's law	T1, R1
15		Crystal structure determination by Laue method	T1, R1

16		Crystal structure determination by Laue and Powder method	T1, R1

7. Lecture Notes

7.1 Introduction:

Solid:

- It is one of the states of matter in which large number of atoms or molecules or ions are closely packed together.
- The physical structure of a solid and its properties related to the arrangement of atoms or molecules or ions within it.

Crystallography:

- The word "crystallography" derives from the Greek words krystallon = clear ice, and graphine= write
- The study of the geometrical form (i.e., arrangement of atoms in solids) and other physical properties of solids by using X-Rays, electron beams and neutron beams etc. is termed as the science of crystallography.

7.2 Classification of solids:

Every solid element has its own internal structure. The internal structure of solid depends on the internal arrangement of atoms or molecules or ions.

Solid are classified in two categories based on the internal arrangement of atoms or molecules.

1. Crystalline solid or crystals and
2. Amorphous solids or no-crystalline solid.

7.2.1. Crystalline solids or crystals:

Solids that have a definite shape and size are called crystalline solids.

Properties of crystalline solids:

- In Crystal solids, the atoms or molecules are arranged in a regular and periodically in three-dimensional manner.
- If the crystal breaks, the broken pieces also have regular shape.
- They have characteristic geometrical shape.

- Some crystalline solids are anisotropic i.e. the physical properties like thermal conductivity, electrical conductivity, refractive index and mechanical strength of crystalline solids are different along different directions.
- Melting point of crystalline solids is extremely sharp. Because, when the temperature increases, bonds break at the same time. This is due to the strength of all the bonds between different atoms, molecules and ions is equal.
- They are most stable as compared to other solids.
- Examples: i. Metallic crystals: Gold, Silver, Aluminum, etc.

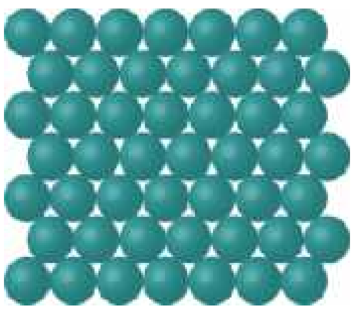
  ii. None-metallic crystals: Diamond, Silicon, Germanium, and Sodium chloride etc.

7.2.2 Amorphous solids or Non-crystalline solids:

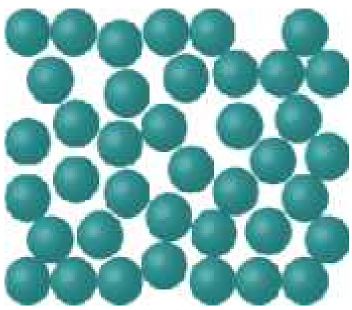
Solids that have no particular external shape are known as amorphous solids. The word amorphous comes from the Greek a means without, and morphé, means shape or form.

7.3 Properties of Amorphous solids:

- In amorphous solids, the atoms or molecules are arranged in an irregular manner.
- If an amorphous solid breaks, the broken pieces are irregular in shape.
- They do not have sharp melting points. This is due to the variable strength of bonds present between the molecules, ions or atoms. So, bonds having low strength on heating break at once. But the strong bonds take some time to break.
- They are isotropic in nature. Isotropic means that in all the directions their physical properties will remain same.
- They are less stable.
- Example Glasses, plastics, Rubbers etc.



Crystalline



Amorphous

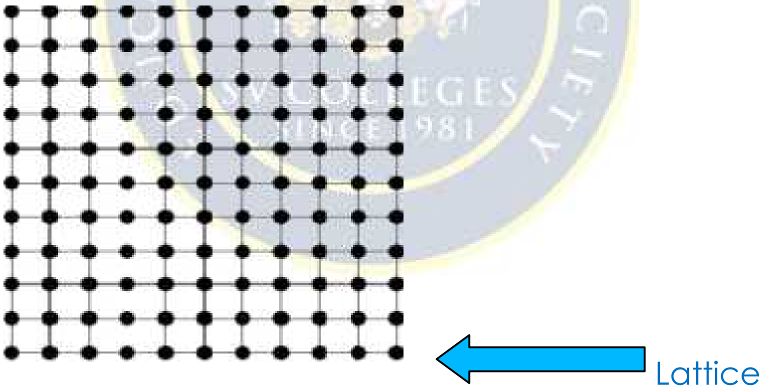
Language of crystals or Crystal geometry:

Crystal geometry is useful to understand the crystal structures. If we want to understand the crystal structures, some basics are needed. Crystal geometry gives us those basics.

7.4 Crystal lattice (or) Space lattice

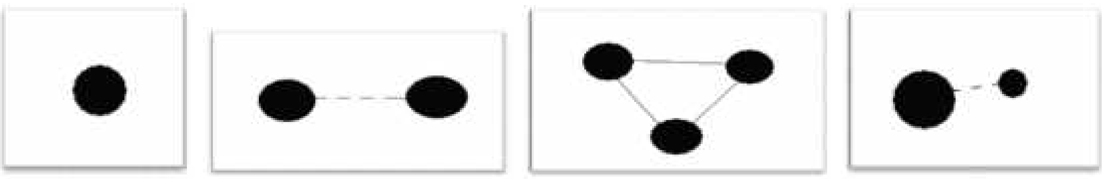
A crystal structure can be studied in terms of an idealized (imaginary) geometrical concept is called a space lattice and is introduced by Auguste Bravias in 1848. According to this concept,

- Each atom in the crystal structure can be replaced by a point in space .These points are known as lattice points.
- A Geometrical representation of the crystal structure in terms of lattice points in three dimensional space is called "space lattice (or) crystal lattice ".
- Lattice points are arranged in regular and periodically in three-dimensional order in space. In which every lattice point has the same environment with respect to all other points.
- Lattice points denote the position of atoms (or) molecules or ions in the crystal (fig.1)
- It is an imaginary concept.



7.4.1. Basis (or) motif:

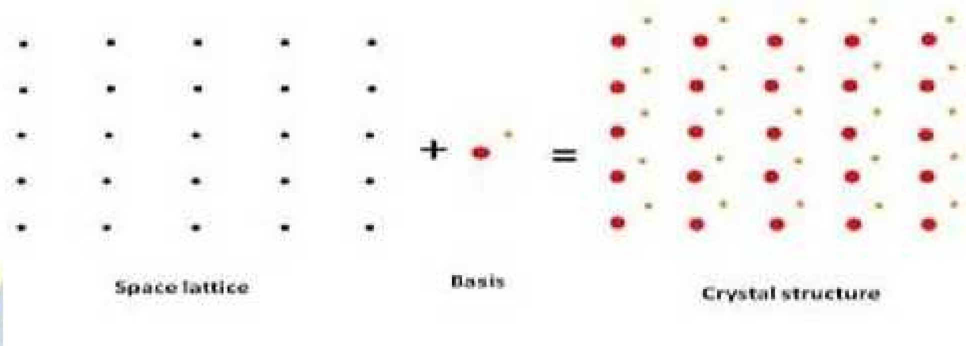
- A Group of atoms or molecules or ions is called 'Basis'
- Basis may consist of single atom or a group of atoms.

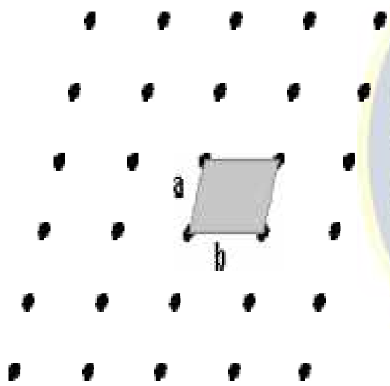


(a) Single atom (b) Di atomic molecule (c) Tri atomic molecule (d) Ion

7.5. Crystal structure:

- Crystal structure is a combination of lattice and basis.
- A crystal structure is formed by adding basis (atoms) to every lattice point of the space lattice. The number of atoms in the basis may be one or more than one.
- Thus the crystal structure is real and the crystal lattice is imaginary
- The logical relation is Crystal structure = Space lattice + basis.





7.5.1. Unit cell:

- Crystal structure can be defined in terms of unit cell.
- The unit cell is the smallest block or geometric figure of crystal, from which the entire crystal is built up by repetition in three dimensional manners.
- It represents the actual structure of crystal.
- The three dimensional representation of unit cell is as shown in figure.

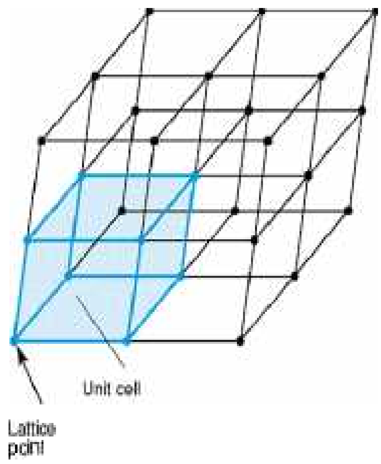


Fig. Unit cell in 2D -space lattice **Fig. Unit cell in 3D -space lattice**

7.5.2 Types of unit cells:

There are two distinct types of unit cells: **primitive** and **non-primitive**.

1. Primitive cell:

- A primitive cell is the simplest type of unit cell which contains only one lattice point per unit cell (contains lattice points at its corner only).
- Primitive unit cells contain only one lattice point, which is made up from the lattice points at each of the corners.
  Ex: Simple cubic (SC)

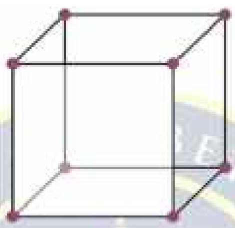
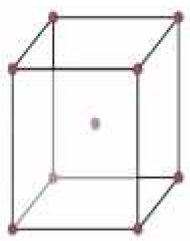


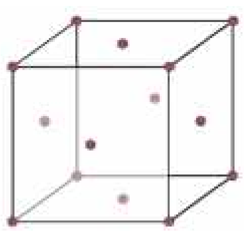
Fig: Simple Cube

2. Non-primitive cell:

If there are more than one lattice points in a unit cell, it is called a Non-primitive cell.

Non-primitive unit cells contain additional lattice points, either on a face of the unit cell or within the unit cell, and so have more than one lattice point per unit cell. Ex: BCC and FCC contain more than one lattice point per unit cell





(a) BCC (b) FCC

7.6 Lattice parameters of unit cell:

Definition:

To represent a lattice unit cell, we require the six parameters i.e., Axial lengths (a,b,c) and interfacial angles (α,β,γ) these quantities are known as "Lattice Parameters".

i.e., Lattice parameters are

1. Axial lengths (a,b,c)
2. Interfacial angles (α, β, γ)

Explanation:

Consider a cubic unit cell with crystallographic axes X, Y, Z as shown in figure.

Let OA=a, OB=b, and OC=c be the intercepts made by the unit cell along the crystallographic axes. These quantities a, b and c are called translational vectors (or) axial lengths or primitives.

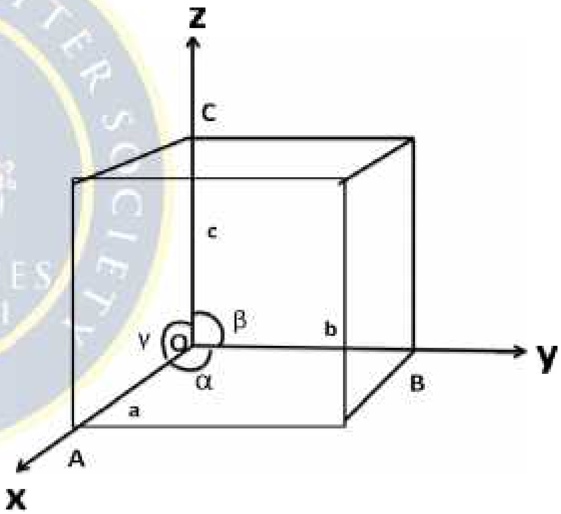


Fig: Cubic unit cell

The angles between the three crystallographic axes are known as interfacial angles.

The angle between a and b is alpha (α) and b and c, is beta (β), and that between a and c is gamma (γ).

These three angles (α, β, γ) are called interfacial angles.

The crystal systems and Bravias lattices:

Crystals are classified into 7 crystal systems on the basis of lattice parameters viz:

(i) Axial lengths a,b,c and

(ii) Interfacial (axial) angles α,β,γ.

The 7 basic crystal systems are

1. Cubic
2. Tetragonal
3. Orthorhombic
4. Monoclinic
5. Triclinic
6. Rhombohedral OR Trigonal
7. Hexagonal

7.7 Bravais Lattices:

Bravais in 1880 showed that there are 14 possible types of space lattices in the 7 crystal systems as shown in table.

According to Bravais, there are only 14 possible ways of arranging points in space lattice from the 7 crystal systems .Theses 14 space lattices are called the Bravais lattices.

Name	**Axial lengths**	**Interfacial angels**	**No.of Bravias lattices**	**Bravias lattice**
1.Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90°$	3	P, I ,F
2.Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$	2	P, I
3.Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$	4	P, I, F, Base
4.Monoclinic	$a \neq b \neq c$	$\alpha = \beta = 90° \neq \gamma$	2	P, Base
5.Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$	1	P
6.Rhombohedral	$a = b = c$	$\alpha = \beta = \gamma \neq 90°$	1	P
7.Hexagonal	$a = b \neq c$	$\alpha = \beta = 90°, \gamma =$	1	P

		120°		

The 7 crystal systems and Bravias lattices are discussed briefly one by one as follows

1. Cubic crystal system:

In cubic crystal system, the three crystal axes perpendicular to each other and axial lengths are the same along the entire three axes as shown fig.

Lattice parameters:

All three sides equal, a=b=c

All three right angles, α= β= γ=900

Fig: cubic system

Examples: Pb, Hg, Ag, Po, Au, Cu, ZnS, diamond, KCl, CsCl , NaCl, Cu2O, CaF2 and alums etc.

Possible Bravais lattices:

Primitive

Body centered

Face centered.

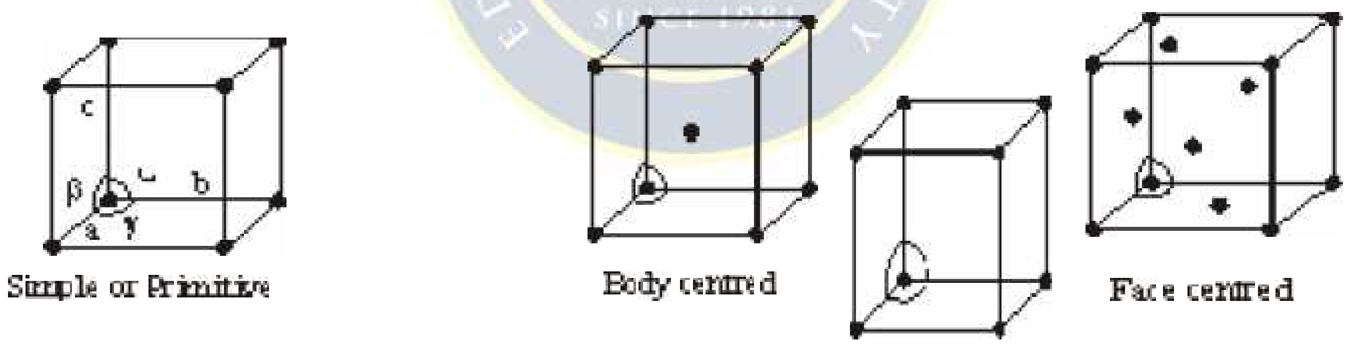
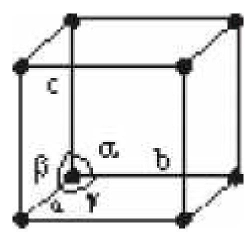


Fig: Possible Bravias lattices of cubic system

2. Tetragonal System:

In tetragonal system, the three crystal axes are perpendicular to each other. Two of the three axis lengths are the same, but the third length is different, as shown in fig



Lattice parameters:

Two sides equal , a=b ≠c

All three right angles; α= β= γ=900

Examples: Niso Sno2 and Indium

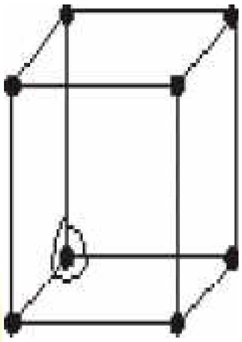
Fig: Tetragonal

Possible Bravais lattices:

- Primitive
- Body centered

Fig: Possible Bravias lattices of tetragonal system

3. Orthorhombic Crystal System:



In orthorhombic crystal system, the crystal axes are perpendicular to each other and all the three axial lengths are of unequal lengths (different), as shown in fig.

Lattice parameters:

- All the sides different a≠b≠c
- All three right angles; α= β= γ=900

Examples: KNO3, Baso4and Mgso4 etc

Possible Bravais lattices:

- Primitive
- Body centered
- Base centered
- Face centered.

Fig: Orthorhombic

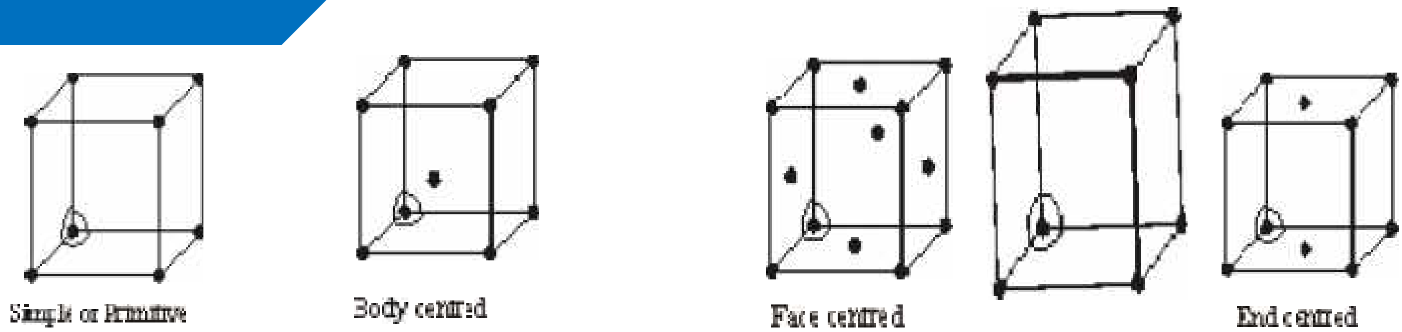


Fig: Possible Bravias lattices of orthorhombic system

4. Monoclinic Crystal system:

In monoclinic crystal system, two of the crystal axes perpendicular to each other, but the third obliquely inclined. The three axial lengths are different along the axes as shown in Fig.

Lattice Parameters:

- All the sides different a≠b≠c
- Two right angles, third arbitrary α=β=900, γ ≠900

Examples: Na2sO4, Feso4 , No2So3 etc

Fig: Monoclinic system

Possible Bravais lattices:

- Primitive
- Base centered

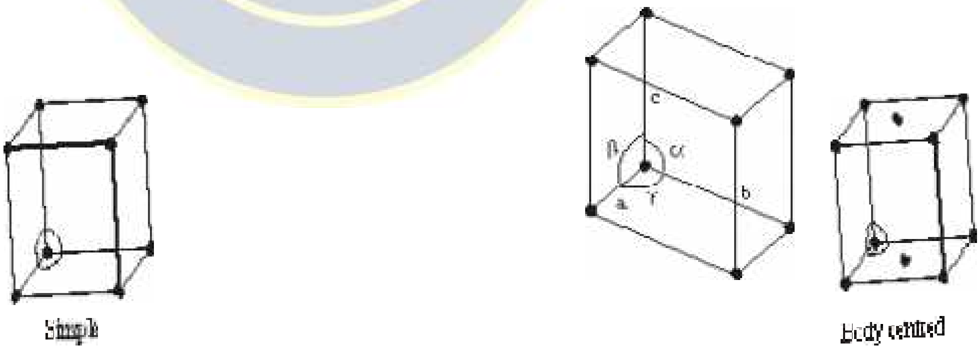


Fig: Possible Bravias lattice of monoclinic system

5. Triclinic crystal system:

In triclinic crystal system, all the three crystal axes are not perpendicular to each other. The axial lengths are also not equal (different) along the three axis, as shown in fig.

Lattice parameters:

- All three sides different a≠b≠c
- All three angles different α≠ β ≠ γ≠900

Examples: Cuso4 and k2cr2o7

Possible Bravais lattices:

- Primitive

Fig: Triclinic system

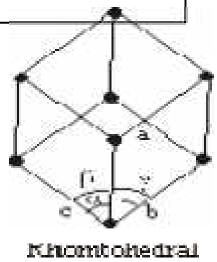
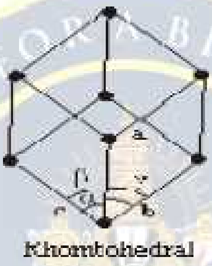


Fig: Possible Bravias lattice of triclinic system

6. Trigonal (Rhombohendral) crystal system:



In trigonal crystal system, the three axes are inclined to each other at an angle other 900. The three axial lengths are equal along three axes as shown in fig.

Lattice parameters:

- All the sides equal a=b=c
- All three angles equal, of arbitrary value α=β≠900

Examples: Caso4, Bi, Sb, calcite etc.

Possible Bravais lattice:

Fig: Trigonal system

- Primitive

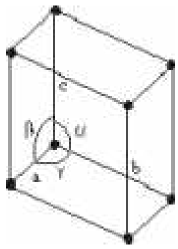


Fig: Possible Bravias lattice of Rhombohedral

7. Hexagonal system:

In hexagonal crystal system, two of the crystal axes are 900 apart, while the third is perpendicular to both of them.

The axial lengths are the same along three axes that are 900 apart, but the axial length along the third axis is different as shown in figure.

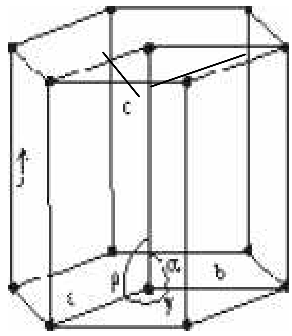
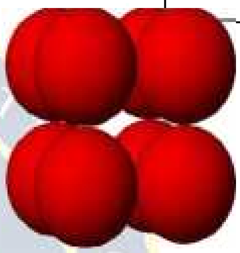


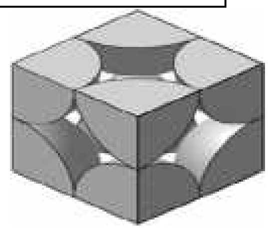
Fig: Hexagonal system

Lattice parameters:

- Two sides equal, third arbitrary a=b≠c
- Two right angles, third angles 1200, α=β=900 ; γ=1200

Examples: Tourmaline, Quartz, AgI and Sio2





Possible Bravais lattices:

- Primitive

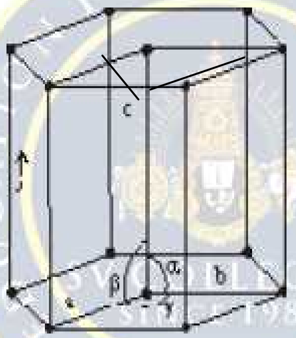


Fig: possible Bravias lattice of hexagonal system

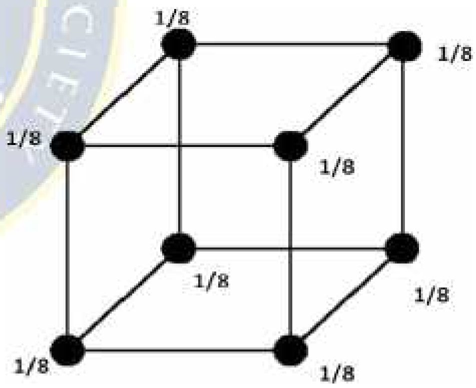


Fig: Simple cubic structure

7.8 Structures of cubic system:

There are three types of structures possible in this system depending on the position of the lattice point (atoms) in the unit cell fig.

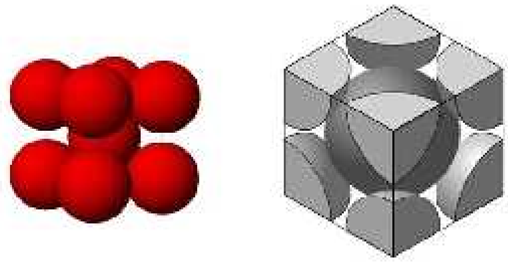
(1) Simple cubic (sc) (or) primitive

(2) Body centered cubic (BCC) and

(3) Face centered cubic (FCC)

7.8.1 Simple cubic (SC) crystal structure:

- In this structure; there are only 8 atoms one at each corner of the cube.
- The corner atoms touch each other along the edges as shown in fig.
- Each and every corner atom is shared by 8 adjacent unit cells.



7.8.2 Body-centered cubic structure (BCC)

- In this case, we have two types of atoms fig,(i) Corner atoms and (ii) Body centered atoms i.e., there are 8 corner atoms, one at each corner of the unit cell and one body centered atom at the centre of the unit cell as shown in fig.
- In this structure, the corner atoms do not touch each other. But each corner atom touches the body centered atom along the body diagonal as shown in fig.

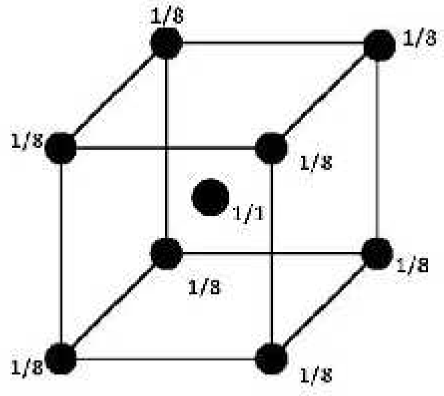


Fig: Body centered cubic structure

- Each and every corner atom is shared by 8 adjacent unit cells and the body centered atom is shared by that particular unit cell alone and is not shared by any other unit cell.

7.8.3 Face-centered cubic structure (FCC):

- In this case, we have two types of atoms viz (i) Corner atoms and (ii) Face centered atoms, i.e. there are 8 corner atoms, one at each corner of the unit cell and six atoms at the centers of six faces of unit cell as shown in Fig.
- In this structure, the corner atoms do not touch each other. But each corner atom touches the face centered atoms along the diagonal of the face of the cube as shown in fig.
- Each and every corner atom is shared by adjacent unit cells, and each face centered atom is shared by only two unit cell, which lie on either side of the atom.

Fig: Face-centered cubic structure (FCC):

7.9 Parameters determine the crystal structure of materials:

Let is discuss some of the important parameters which are used to describe the crystal structure.

1. Number of atoms per unit cell (or) Effective number:

The total number of atoms present in (or) shared by an unit cell is known as number of atoms per unit cell.

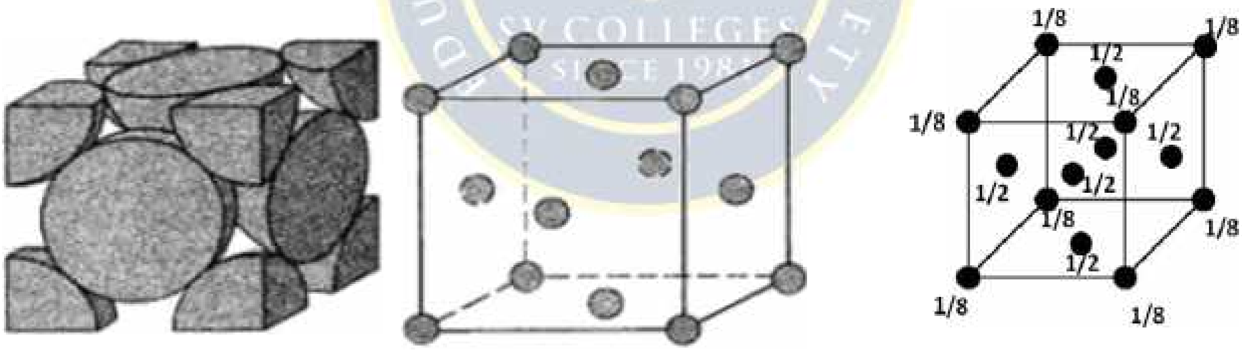
2. Co-ordination number:

It is the number of nearest neighboring atoms to a particular atom.

3. Atomic radius:

Atomic radius is defined as half of the distance between any two nearest neighbor which have direct contact with each other, in a crystal of pure element. It is usually expressed in terms of cube edge ‘a’.

4. Atomic packing factor (packing density):



Atomic packing factor is defined as the ratio between the volume occupied by the total number of atoms per unit cell (v) to the total volume of the unit cell (V).

i.e. $APF = \dfrac{\text{Volume occupied by the total no.of atoms per unitcell}}{\text{Totalvolume of the unit cell}} = v/V$

$= \dfrac{\text{No.of atoms per unit cell x volume of oneatom}}{\text{Total volume of the unit cell}}$

5. Void space or interstitial space:

The void space in the unit cell is the vacant space left unutilized in the unit cell. It is equal to (1-APF).It is often expressed as percentage.

$$\text{Void space} = (1-\text{APF}) \times 100.$$

6. Density of solid:

The density of a crystalline solid is defined as the ratio of the unit cell and the volume of a unit cell.

$$\text{Density } (P) = \frac{\text{Mass of unit cell}}{\text{volume of unit cell}}$$

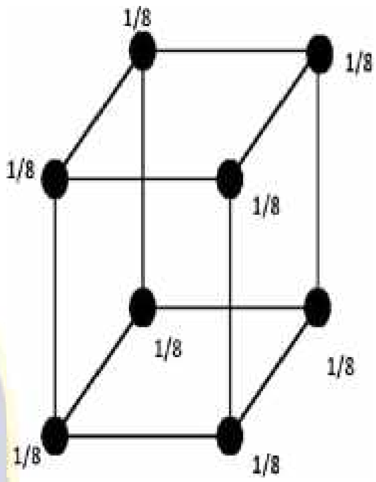


Fig. Simple cubic structure

Let us discuss all the above parameters one by one for a simple cubic structure.

Number of atoms per unit cell:

Definition:
The total number of atoms present in (or) shared by a unit cell is known as number of atoms per unit cell.

Note: This number depends on the number of corner atoms, body centered atom face cantered atom, which varies from structure to structure.

Let us evaluate the number of atoms per unit cell for the three systems.

(a)Simple cubic structure:

- We know that, In simple cubic structure, there are only 8 atoms, one at each corner of the cube (or) the unit cell and each and every corner atom is shared by 8 adjacent unit cells as shown in fig.

i.e., total number of atoms per unit cell = $\frac{1}{8}$ x total number of corner atoms

$$= \frac{1}{8} \times 8 = 1$$

Therefore, the number of atoms per unit cell in S.C is one. Thus, simple cubic is a primitive unit cell.

(b).Body centered cubic structure:

- In this case, we have two types of atoms namely
  (i) Corner atoms (ii) Body centered atoms
- There are 8 corner atoms, one at each other corner of the unit cell and one body centered atom at the centre of the unit cell as shown in fig.

(I)Number of corner atoms per unit cell

Each and every corner atom is shared by 8 adjacent unit cells.

The total number of corner atoms per unit cell = $\frac{1}{8}$ x8=1

(ii)Number of body centered atoms per unit cell

The body centered atom is shared by that particular unit cell alone and is not shared by any other unit cell.

The number of body centered atoms per unit cell =1/1 x1=1

Total no. of atoms = Total no. of corner atoms per unit cell in Bcc + Total no. of body centered atoms per unit cell.

=1+1 = 2

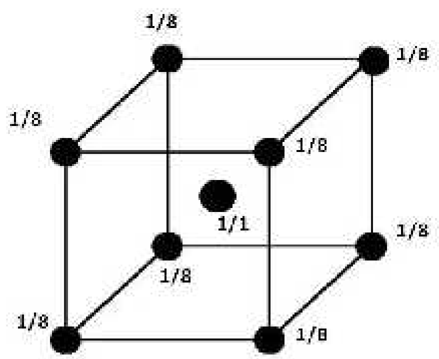


Fig: Body centered cubic structure

Therefore, the number of atoms per unit cell in BCC is two. Thus, BCC is a non- primitive unit cell.

(C).Face-centered cubic structure:

- In this case, we have two types of atoms namely
  (i) Corner atoms (ii) Face centered atoms.
- There are 8 corner atoms, one at each corner of the unit cell and six atoms at the centers of six faces of unit cell as shown in fig.

(i) Number of corner atoms per unit cell

Each and every atom is shared by 8 adjacent unit cells.

The total number of corner atoms per unit cell= $\frac{1}{8}$ x 8=1

(ii) Number of face centered atoms per unit cell

Each face centered atom is shared by only two unit cells, which lie either side of the atom (similarly we have six face centered atoms in an unit cells)

The total number of centered atoms per unit cell

=1/2 x6 = 3

The total no. of atoms per unit cell in FCC= Total no. of corner atoms per unit cell +total no. of face centered atoms per unit cell

=1+3 = 4

Therefore, the number of atoms per unit cell in FCC is four. Thus, FCC is a non- primitive unit cell.

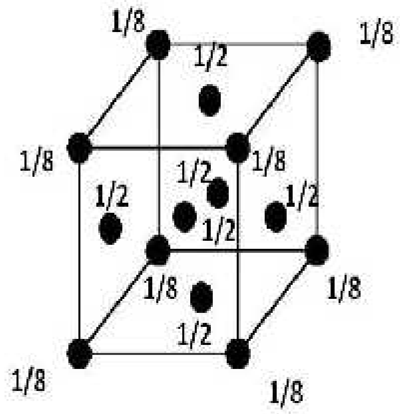


Fig :Face-centered cubic structure (FCC)

Coordination Number:

Definition:

Coordination Number is the number of nearest neighboring atoms to a particular atom which are direct contact with each other.

The Coordination Number for the three types of cubic crystal structure can be calculated as follows.

(a).Simple Cubic Structure:

- In this case, there are only 8 atoms, one at each corner of the cube (or) unit cell. For a particular atom say 'C' atom, there are 4 nearest neighboring atoms, i.e. atoms 1,2,3 and 4 on its own plane and there are 2 more nearest atoms i.e., atom-5 directly above the plane and atom-6 directly below the plane as shown in figure.

- Therefore,The total Number of neighboring atoms to particular atom (C)

=4+1+1=6

Hence, the coordination for SC =6

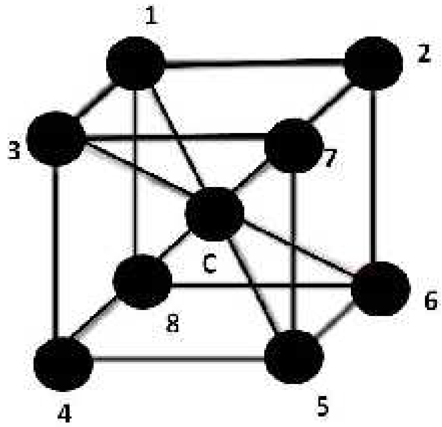


Fig: Co- ordination number in Simple cubic structure

(b). Body centered cubic Structure:

- In this case, we have two types of atoms namely 1.Corner atoms 2. Body centered atoms i.e., there are 8 corner atoms, one at each corner of the unit cell and one body centered atom at the center of the unit cell as shown in figure.
- The corner atoms do not touch each other. But each corner atom touches the body centered along the body diagonal. Thus for particular atom 'C' at the body centre obviously, there are 8 nearest neighbor (corner atoms).

Hence, the coordination for BCC =8

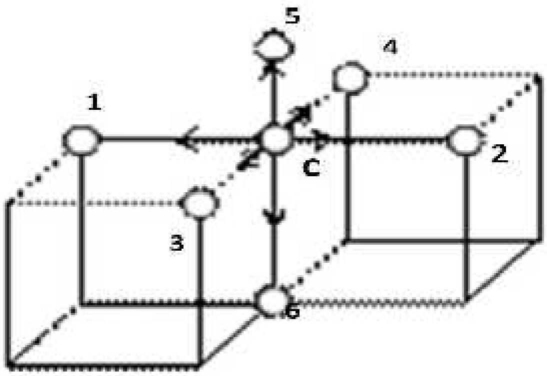


Fig: Co- ordination number in BCC

(C). Face Centered Cubic Structure (FCC):

- In this case, we have 2 types of atoms namely, 1.Corner atoms and 2. Face centered atoms. There are 8 corner atoms, one at each corner of the unit cell and six atoms at the centers of six faces of unit cells as shown in figure.
- Let us consider 2 unit cells one above the other if the reference atom 'C' taken as the face centered atom then it is surrounded by 4 corner atoms on the plane , and 4-face centered atoms above the plane and 4-face centered atoms below the plane as shown figure.
- Therefore the coordination number for FCC=4+4+4=12

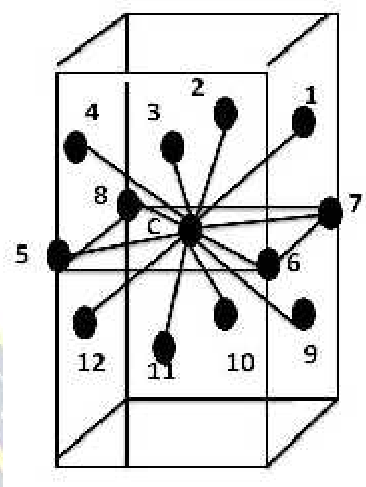


Fig: Co- ordination number in FCC

Atomic Radius:

Definition:

Atomic radius is defined as half of the distance between any two nearest neighbour atoms which have direct contact with each other.

It can be expressed in terms of cube edge a and vice versa.

All the atoms are assumed to be spherical in shape.

For the three cubic structures it can be calculated as follows.

(a).Simple Cubic Structure:

- In simple cubic (SC) structure the corner atoms touch other along the edges as shown in fig.

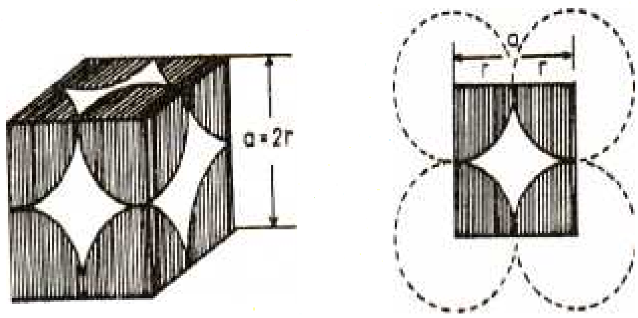


Fig: Simple cubic structure

- Let us consider one of face of the simple cubic structure as shown in fig (2).

Hence the nearest neighbor distance is 2r = a

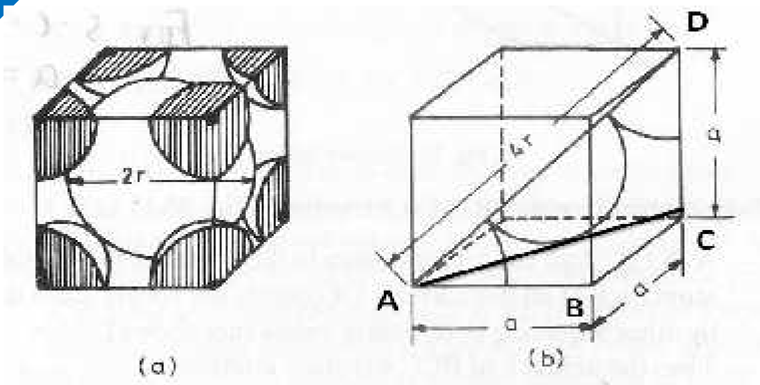


Fig :Body centered cubic structure (BCC)

(b).Body centered cubic structure (BCC):

In BCC structure, the corner atoms do not touch each other. But each corner atom touches the body centered atom along the body diagonal as shown in fig

Therefore the two corner atoms (A and D) situated at the opposite ends can be joined by drawing a diagonal as shown in fig. Thus, the diagonal of the cube AD is 4r.

But, from the geometry of the fig; we can write

(AD) 2 = (AC) 2+ (CD) 2

= (AB) 2+ (BC) 2+ (CD) 2

[From fig ; AC2=AB2+BC2 and AB=BC=CD=a]

= a2+ a2+ a2

=3a2

(Or) AD=a$\sqrt{3}$; But AD = 4r

4r = a$\sqrt{3}$

$$r = \frac{a\sqrt{3}}{4}$$

$$\boxed{\therefore \textbf{ Atomic radius r} = \frac{a\sqrt{3}}{4}}$$

(C) Face-centered cubic structure (FCC):

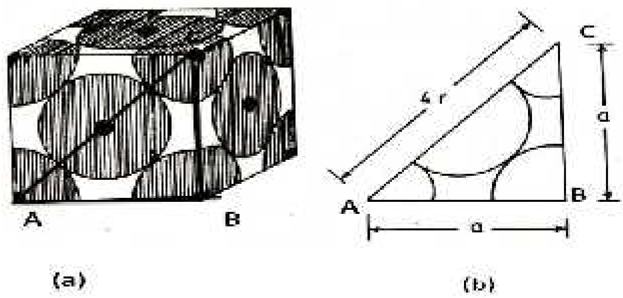


Fig :Face centered cubic structure (FCC)

In FCC structure the corner atoms do not touch each other. But each corner atom touches the face centered atoms along the diagonal of the face of the cube as shown in fig.

Therefore, the two corner atoms (A and C) situated at the opposite ends of the same face can be joined by drawing a diagonal a shown in fig. Thus, the diagonal of the cube AC=4r.

From the geometry of the figure, we can write

$$(AC)^2 = (AB)^2 + (BC)^2$$

$$= a^2 + a^2$$

$$AC^2 = 2a^2$$

$$AC = a\sqrt{2}\text{ ; But } AC = 4r$$

$$4r = a\sqrt{2}$$

$$r = \frac{a\sqrt{2}}{4}$$

$$\boxed{\therefore \text{ Atomic radius } r = \frac{a\sqrt{2}}{4}}$$

4. Atomic packing factor (APF) or packing density:

Definition:

Atomic packing factor is defined as the ratio of the volume occupied by the total number of atoms per unit cell to the total volume occupied by the unit cell.

i.e., APF = $\dfrac{\text{Volume occupied by the total no.of atoms per unitcell}}{\text{Totalvolume of the unit cell}}$ = v/V

= $\dfrac{\text{No.of atoms per unit cell x volume of oneatom}}{\text{Total volume of the unit cell}}$

The packing factor (or) packing density of the three cubic systems can be calculated as follows.

(a).Simple cubic structure (SC):

In simple cubic,

The number of atoms per unit cell=1

Volume of 1 atom (spherical) = $\frac{4}{3}\pi r^3$

$\therefore$ Volume occupied by the total no. of atoms per unit cell (v)

= No. of atoms per unit cell x volume of one atom

= 1x $\frac{4}{3}\pi r^3$

We know the radius of atom in simple cubic $r = \frac{a}{2}$

=1x1x $\frac{4}{3}\pi \left(\frac{a}{2}\right)^3$

Volume of the unit cell (V) = length x breadth x height

We know that for a cubic system, length=breath=height=a

∴ V=a x a x a

= a3

∴ A.P.F = v/V = $\frac{1\times\frac{4}{3}\pi\left(\frac{a}{2}\right)^3}{a^3}$

= $\frac{\pi}{6}$ = 0.52

$$\boxed{\therefore APF = \frac{\pi}{6} = 0.52}$$

And Void space = (1-APF) x 100

= (1-0.052) x 100

=48%

Therefore, we can say that 52% volume of the unit cell of sc is occupied by atoms and remaining 48% volume is vacant. Thus, the packing density 52%

Since the packing density is very low, SC has loosely packed structure.

(b)Body-centered cubic Structure:

In body-centered cubic structure,

The number of atoms per unit cell=2

Volume of one atom (spherical) = $\frac{4}{3}\pi r^3$

∴ Volume occupied by the total no. of atoms per unit cell (v) =

No.of atoms per unit cell x Volume of one atom

=2 x $\frac{4}{3}\pi r^3$

=2 x $\frac{4}{3}\pi[a\sqrt{3}/4]^3$

= $\frac{8\pi}{3}\left[\frac{a^3 \times 3\sqrt{3}}{4\times4\times4}\right]$

v = $\pi a^3\sqrt{3}/8$

Volume of the unit cell for a cubic system (V) =a3

∴APF = $\frac{v}{V}$ = $\pi a^3\sqrt{3}/8/a^3$

$$=\frac{\pi\sqrt{3}}{8}$$

$$=0.68$$

$$\boxed{\therefore\ \mathbf{APF}=\frac{\pi\sqrt{3}}{8}=\mathbf{0.68}}$$

And Void space = (1-APF) x 100

= (1-0.068) x 100

=32%

Therefore, we can say that 68% volume of the unit cell of BCC is occupied by atoms and remaining 32% volume is vacant. Thus, the packing density 68%.

Since the packing density is very low, SC has loosely packed structure.

(C) Face-centered cubic structure:

In face centered cubic structure,

The number of atoms per unit cell=4

Volume of one atom (spherical) $=\frac{4}{3}\pi r^3$

$\therefore$ Volume occupied by the total no.of atoms per unit cell (V) =No.Of atoms perunit cell x volume of one atom

$$=4\times\frac{4}{3}\pi r^3$$

We know that the radius of atom in FCC is $r=\frac{a\sqrt{2}}{4}$

$$\therefore V=4\times\frac{4}{3}\pi\left(\frac{a\sqrt{2}}{4}\right)^3$$

$$=\pi a^3\sqrt{2}/6$$

Volume of the unit cell for a cubic system (V) $=a^3$

$$\therefore APF=\frac{v}{V}=\frac{\frac{\pi a^3\sqrt{2}}{6}}{a^3}$$

$$=\pi\frac{\sqrt{2}}{6}$$

$$=0.74$$

$$\therefore \text{ APF} = \frac{\pi\sqrt{2}}{6} = 0.74$$

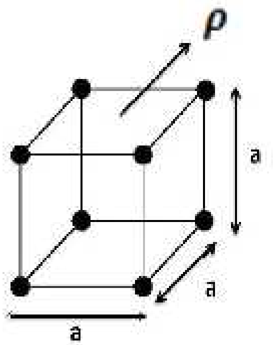


Fig:Cubic unit cell

And Void space = (1-APF) x 100

= (1-0.074) x 100

=26%

Therefore, we can say that 74% volume of the unit cell of FCC is occupied by atoms and remaining 26% volume is vacant. Thus, the packing density 74% Since the packing density is very high, FCC has closely packed structure.

Density of Crystalline solid

The density of a crystalline solid is defined as the ratio of mass of the unit cell and the volume of a unit cell.

$$\text{Density (P)} = \frac{\text{Mass of unit cell}}{\text{Volume of unit cell}}$$

Let us consider a cubic unit cell of the following parameters.

- The lattice constant of the cell=a
- The number of atoms per unit cell=n
- The atomic weight of crystalline substance=M
- The density of unit cell= ρ
- The Avogadro's number=N
- The volume of unit cell=a3

$$\text{Mass of each atom in unit cell} = \frac{M}{N}$$

If there are 'n' no. of atoms in a unit cell, then the mass of the cubic unit cell = $\frac{nM}{N}$

$$\rho = \frac{\frac{nM}{N}}{a3} = \frac{nM}{Na3}$$

$$\therefore \rho = \frac{nM}{Na3}$$

And $a3 = \frac{nM}{N\rho}$

$$a = \left(\frac{nM}{N\rho}\right)^{1/3}$$

The above expression represents the expression for lattice constant.

The density for the three types of cubic crystal structure can be calculated as follows,

(a) Simple Cubic structure (SC):

For SC; n=1

$$\therefore \rho = \frac{M}{Na3}$$

(b) Body-centered cube (BCC)

For BCC n=2

$$\therefore \rho = \frac{2M}{Na3}$$

(C) Face centered cubic Structure (FCC)

For FCC; n=4

$$\therefore \rho = \frac{4M}{Na3}$$

7.10 Crystal Planes or Lattice Planes:

We know that, in lattice atoms or molecules or ions are arranged in regular and periodically in three-dimensional order in a space. In which every lattice point has the same environment with respect to all other points.

If we draw a number of parallel planes through these lattice points, then they are known as lattice planes or crystal planes.

Therefore, crystals are made up of large number of parallel equidistant planes, passing through the lattices points called crystal planes.

The representation of lattices planes in different ways as shown in fig.

The problem is that how to designate these planes in a crystal. Miller introduced a method to designate a set of parallel planes in a crystal by three numbers (h k l) known as Miller indices.

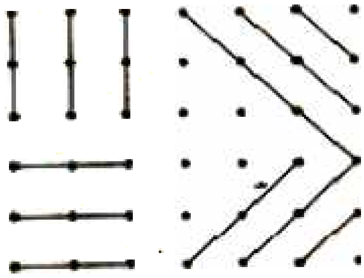
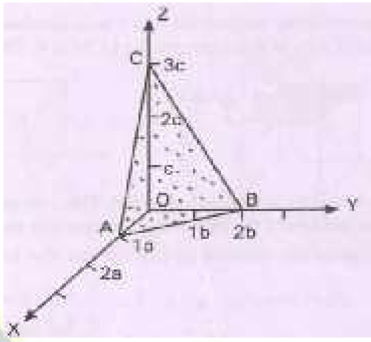


Fig: The representation of lattices planes in different ways.

7.10.1 Miller indices:



- Miller introduced to a set of three numbers to designate plane in a crystal. This set of three numbers is known as Miller indices of the concerned plane and is denoted by (h k l).
- Miller indices are the reciprocals of the intercepts made by the plane along the three crystallographic axes which are reduced to smallest numbers.

Produce for finding Miller indices:

1. Find the intercepts made by the plane along the three crystallographic axes.
2. Express the intercepts in terms of multiples of axial lengths.
3. Find the reciprocal of the numerical intercept values.
4. Convert these reciprocals into whole numbers by multiplying each with their LCM.
5. Enclose these numbers in bracket (→).This represents the Miller indices of the concerned plane

Example:

Step 1: Find the intercepts made by the plane along the three crystallographic axes (p, q, r).

Let us consider the plane ABC along the three axes X, Y, and Z, which cuts 1- unit along the X-axis, 2-units along the Y-axis and 3-units along the Z-axis as shown in fig.

From fig; Intercepts p = 1, q =2 and r =3

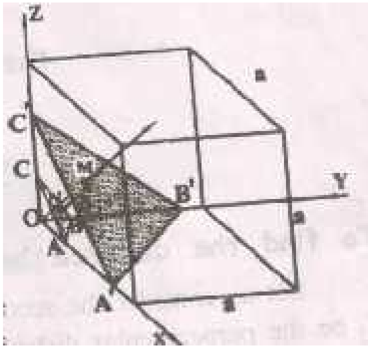
Step2: Express the intercepts in terms of multiples of axial lengths (pa, qb, rc).

From fig; Pa: qb: rc

1a: 2b: 3c

Where p = 1, q =2 and r =3

Step 3: Find the reciprocal of the numerical intercept values ($\frac{1}{p}, \frac{1}{q}, \frac{1}{r}$)



$$\frac{1}{p}:\frac{1}{q}:\frac{1}{r}$$

$$\frac{1}{1}:\frac{1}{2}:\frac{1}{3}$$

Step4: Convert these reciprocals into whole numbers by multiplying each with their LCM.

$$\frac{1}{1}X6:\frac{1}{2}X6:\frac{1}{3}X6$$

6:3:2

Step5: Enclose these numbers in bracket $\rightarrow$ (6 3 2). This represents the Miller indices of the concerned plane

7.11 Interplaner distance or spacing (d):

The separation between two successive or adjacent lattice planes is called interplaner distance or spacing and it is denoted by'd'.

Determination of Interplaner distance between two successive Planes.

- Consider two successive lattice planes ABC and AIBICI in a cubic unit cell as shown in fig 1.
- Let ON = d1 be the perpendicular distance or normal between the origin (O) and the first plane ABC.
- Let OM = d2 be the perpendicular distance or normal between the origin (O) and the first plane ABC and
- Let MN =d be the interplaner distance between two planes ABC AND AIBICI.

$$\text{i.e., } MN = OM - ON$$
$$d = d_2 - d_1$$

Fig: Planes in a cubic unit cell

Calculation of d1:

- Let us consider the plane ABC as shown in fig.
- Let ON = d1 be the perpendicular distance or normal between the origin (O) and the first plane ABC.
- Let α, β and γ be the angles between ON and X,Y,and Z axes respectively .
- Let OA = $\frac{a}{h}$, OB = $\frac{b}{k}$ and OC = $\frac{c}{l}$ be the intercepts made by the orthogonal axis OX, OY and OZ.

- From ΔONA

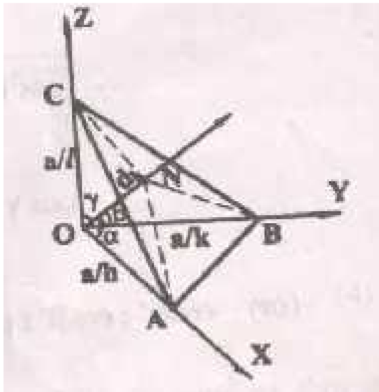
$$\text{Cos } \alpha = \frac{ON}{OA} = \frac{d_1}{\frac{a}{h}} = \frac{d_1 h}{a} \longrightarrow (1)$$

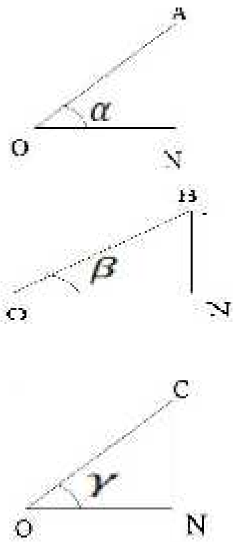
- From ΔONB

$$\text{Cos } \beta = \frac{ON}{OB} = \frac{d_1}{\frac{b}{k}} = \frac{d_1 k}{b} \longrightarrow (2)$$

- From ΔONC

$$\text{Cos } \gamma = \frac{ON}{OC} = \frac{d_1}{\frac{c}{l}} = \frac{d_1 l}{c} \longrightarrow (3)$$





According to the law of directions of cosine law

$$\text{Cos}^2 \alpha + \text{Cos}^2 \beta + \text{Cos}^2 \gamma = 1 \longrightarrow (4)$$

Substituting equations (1) (2) & (3) in equation (4)

$$\frac{d_1^2 h^2}{a^2} + \frac{d_1^2 k}{b^2} + \frac{d_1^2 l}{c^2} = 1$$

For cubic system a=b=c;

$$\frac{d_1^2 h^2}{a^2} + \frac{d_1^2 k}{a^2} + \frac{d_1^2 l}{a^2} = 1$$

$$\frac{d_1^2}{a^2}(h^2 + k^2 + l^2) = 1$$

$$d_1^2 = \frac{a^2}{h^2+k^2+l^2}$$

$$d_1 = \frac{a}{\sqrt{h^2+k^2+l^2}} \longrightarrow (5)$$

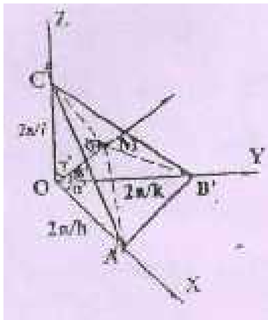
This is the perpendicular distance from origin to the first plane

Calculation of d2:

- Let us consider the plane AIBICI as shown in fig.
- Let OM = d2 be the perpendicular distance or normal between the origin (O) and the first plane AIBICI.
- Let αI, βI and γI be the angles between ON and X,Y,and Z axes respectively .
- Let OAI = $\frac{2a}{h}$, OBI = $\frac{2h}{k}$ and OCI = $\frac{2c}{l}$ be the intercepts made by the orthogonal axis OX, OY and OZ.
- From fig, the perpendicular distance from origin to the second plane Can be calculated as

$$\boxed{d_2 = \frac{2a}{\sqrt{h^2+k^2+l^2}}} \longrightarrow (6)$$

- Therefore, the interplaner distance between two planes ABC AND AIBICI is d = d2-d1



$$d = \frac{2a}{\sqrt{h^2+k^2+l^2}} - \frac{a}{\sqrt{h^2+k^2+l^2}}$$

$$\boxed{d = \frac{a}{\sqrt{h^2+k^2+l^2}}}$$

The above equation represents the interplaner distance between any two successive planes in a cubic crystal.

7.12 X-ray diffraction:

- Like light rays X-rays also exhibit diffraction phenomena when they passes through the crystals.

- The bending of X-rays around the obstacle. X-rays are mostly used to study the crystal structures because the wavelength of X-rays (~ 1A0) is comparable with the interatomic distance or spacing. This was first discovered by the German physicist Max von Laue in 1912.
- According to Max von Laue, a crystal serves as a natural three dimensional grating for X-rays, where the regular and periodically in three-dimensionally arrangement of atoms or molecule or ions act as the parallel lines or grating elements in the 2D optical grating.
- His associates, W.Fredarich and P.Knipping proved experimentally that X-rays are waves and that the atoms/molecules/ions in a crystal are arranged in regular manner.
- In the same year, W.H.Bragg and W.L.Bragg derived a relation between the wavelength of X-rays, angle of diffraction and separation of atomic planes in the crystal called Braggs law.

7.13 Braggs Law:

In 1972, W.H.Bragg and W.L.Bragg derived a relation between the wavelength of X-rays(λ), angle of diffraction(θ) and separation of atomic planes(d) in the crystal called Braggs law.

$$\boxed{\text{i.e., } 2d\sin\theta = n\lambda}$$

Derivation :

- Let us consider a crystal consists of equidistant and parallel planes with the interplaner distance (d) as shown in fig.
- When the X-rays incident on the atoms in the planes and diffract the x-rays in all directions.
- Let the X-ray AB is incident on the atom B in the plane 1 with an angle θ and diffracted in the direction of BC.
- Let another X-ray DE is incident on the atom E in the plane 2 with an angle θ and diffracted in the direction of EF.
- These diffracted rays will superimpose with each other depending on the path difference between the X-rays.

- To calculate the path difference, two normal's BF and BG from B to BF and BG. is shown in fig.

Fig : Diffraction of X-rays in the planes

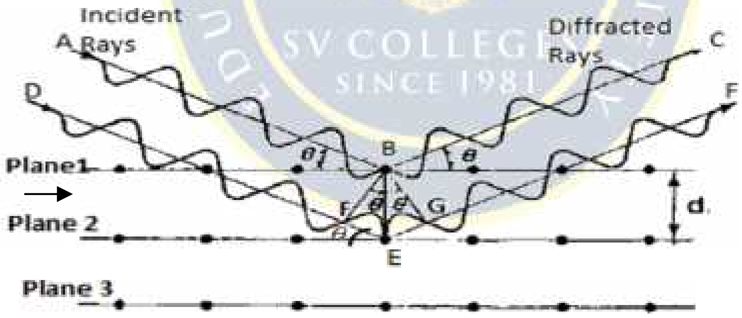
- From Fig,Δ = FE +EG $\longrightarrow$(1)
- From ΔBFE

$$\text{Sin}\theta = \frac{FE}{BE} = \frac{FE}{d}$$

FE= d Sinθ $\longrightarrow$ (2)

- From ΔBGE

$$\text{Sin}\theta = \frac{EG}{BE} = \frac{EG}{d}$$



EG= d Sinθ $\longrightarrow$ (3)

Substituting equations (2) and (3) in equation (1), we have

Δ = 2d Sinθ $\longrightarrow$ (4)

We know that, the condition for maximum intensity or constructive interference is given by

Δ =nλ $\longrightarrow$ (5)

From the two equations (4) &(5),we get

$$\boxed{2d\sin\theta = n\lambda} \longrightarrow (6)$$

The above equation is known is known as Bragg's equation.

Importance:

- From the Bragg's Law

$$d = \frac{Si\,n\theta}{n\lambda}$$

  Knowing the values of the wavelength of X-rays(λ) and angle of diffraction(θ), separation of atomic planes(d) can be calculated.

- Knowing the value of d, lattice constant 'a' of the cubic crystal can be calculated.
- For cubic crystal; $d = \frac{a}{\sqrt{h^2+k^2+l^2}}$

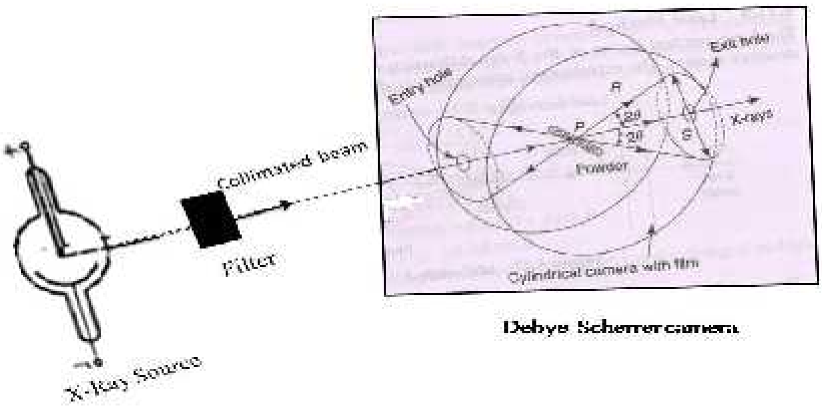
Knowing the value of d, lattice constant 'a', values of $h^2 + k^2 + l^2$ can be calculated and we can classify the SC, BCC and FCC crystal systems.

The Powder Method or Debye –Scherrer Method:

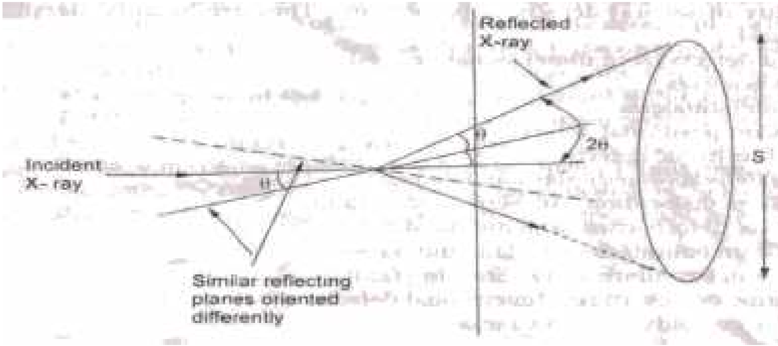
The powder diffraction method was devised independently in 1916 by Debye and Scherrer in Germany and in 1917 by Hull in the United States.

This method is used to study the structure of the polycrystalline solids in the form of a fine powder, containg a large number of tiny crystallites with random orientations.

The experimental arrangement of Debye- Scherrer method is shown in fig.



Debye Scherrer camera



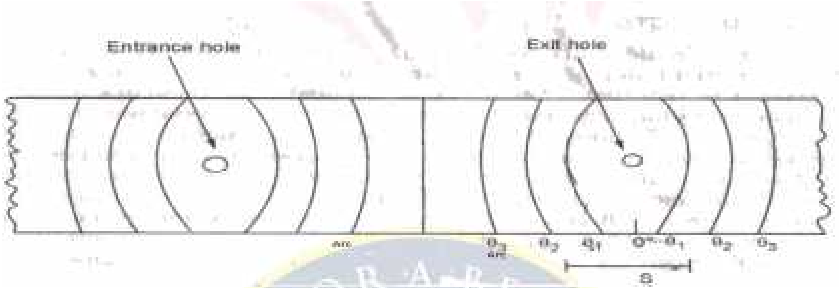


Fig : Photographic Film

- It consists of a source(S) of X-rays.
- The beam of X-rays is monochromatised by passing through a filter and collimated by passing through a collimator.
- The collimator produces collimated beam and falls on the powder sample or specimen.
- The powder sample or specimen is taken a thin walled non diffracting capillary tube.
- The specimen is placed in the Debye-Scherrer camera and the photographic film is mounted or located round the inner surface of the camera covering nearly the whole circumference in order to receive the beams diffracted up to 1800.
- The specimen is suspended vertically on the axis of a Debye-Scherrer camera.
- The collimated X-rays beam passing through the entry hole and falls on the powdered specimen, since crystallites with random orientations, almost all the possible θ and d values are available for diffraction of X-rays.
- The diffraction takes places for the values of θ and d which satisfies the Braggs condition, i.e., $2d\sin\theta = n\lambda$.
- The diffracted X-rays come out through the exit hole in the form of a cone.

- The diffracted X-ray cones make impressions on the film in the form of arcs on either side of the exit and entry holes with their centers coinciding with the holes.
- The film is exposed for a few hours in order to obtain arcs of sufficiently high intensity.
- It is than removed from the camera and developed.
- The arcs are produced on the developed film appear dark, as shown in fig.
- From fig ; the angle θ corresponding to a particular pair of arcs is given by

$$4\theta \text{ (radians)} = \frac{S}{R} \qquad \left(\text{since, angle}=\frac{arc}{radius}\right)$$

Where S is the arc length (i.e., distance between two arcs) and

R is the radius of the camera.

$$4\theta \text{ (degress)} = \frac{S}{R}\left(\frac{180}{\pi}\right) = \frac{57.296\,S}{R}$$

From the above expression, θ can be calculated.

Hence, knowing the values of θ and λ, the inter planer distance d for a set of parallel planes can be calculated using the Braggs condition.

$$d = \frac{\lambda}{\sin\theta}$$

These d values are used to find the space lattice of the crystal structure.

8. Practice Quiz

1.	The smallest part of the crystal is								[d]
	a	Primitive	b	Crystal	c	Basis	d	Unit cell	
2.	Lattice+basis=								[b]
	a	Unit cell	b	Crystal	c	Amorphous	d	none	
3.	The possible number of bravais lattice is								[b]
	a	5	b	14	c	7	d	21	
4.	The number of atoms per unit cell in bcc lattice is								[b]
	a	1	b	2	c	4	d	8	
5.	The coordination number of a simple cubic is								[a]

	a	6	b	8	c	12	d	13	
6.	The coordination number of a bcc is								[b]
	a	6	b	8	c	12	d	13	
7.	The coordination number of a bcc is								[c]
	a	6	b	8	c	12	d	12	
8.	The miller indices of the plane parallel to z-axis are								[c]
	a	(001)	b	(101)	c	(110)	d	(100)	
9.	The coordination number of NACL structure is								[c]
	a	4	b	5	c	6	d	7	
10.	The atomic packing factor for bcc structure is								[b]
	a	0.74	b	0.68	c	1.00	d	0.52	

9. Assignment Questions:

S. No	Question	BL	CO
1	Describe seven crystal system with diagram.	2	5
2	Describe the bravis lattices.	2	5
3	Explain the crystal structure of simple cubic.	2	5
4	Explain the crystal structure of BCC.	2	5
5	Explain the crystal structure of FCC.	2	5
6	Explain the Bragg's law of X-ray diffraction	2	5

10 . Part A- Question & Answers

S. No	Question& Answers	BL	CO
1	Define Crystallography. Answer: The study of the geometrical form and other physical properties c crystalline solids by using X-Ray, electron beams and neutron beams etc. i termed as the science of crystallography.	1	5

2	Define space lattice. Answer: A Geometrical representation of the crystal structure in terms of lattice points is called "space lattice (or) crystal lattice.	1	5
3	What is meant by basis? Answer: A Group of atoms or molecules or ions is called 'Basis'.	1	5
4	Explain the term crystal structure. Answer: A crystal structure is formed by adding basis (atoms) to every lattice point of the space lattice. The number of atoms in the basis may be one or more than one. The logical relation is Crystal structure = Space lattice + basis.	2	5
5	Explain primitive unit cell and give examples. Answer: A primitive cell is the simplest type of unit cell which contains only one lattice point per unit cell (contains lattice points at its corner only). Ex: Simple cubic (SC)	2	5
6	What are interfacial angles? Answer: To represent a lattice unit cell, we require the six parameters i.e., Axial lengths(a,b,c) and interfacial angles (α,β,γ) these quantities are known as "Lattice Parameters.	1	5
7	Explain number of atoms per unit cell and write numbers of atoms for SC, BCC and FCC unit cells. Answer: The total number of atoms present in (or) shared by an unit cell is known as number of atoms per unit cell. The number of atoms per SC unit cell = 1 The number of atoms per BCC unit cell = 2	2	5

	The number of atoms per FCC unit cell = 4		
8	what are the miller indices? Answer: Miller indices are the reciprocals of the intercepts made by the plane along the three crystallographic axes which are reduced to smallest numbers.	1	5
9	Define non- primitive unit cell and give examples. Answer: If there are more than one lattice points in a unit cell, it is called a Non-primitive cell. Ex: BCC and FCC.	1	5
10	What is unit cell? Answer: A unit cell is the smallest block or geometric figure of crystal, from which the entire crystal is built up by repetition in three dimensional manners.	1	5

11. Part B- Question

S. No	Question	BL	CO
1	Explain Crystal systems and their Bravais lattices with neat diagrams.	2	5
2	Define packing factor? Calculate packing factors for SC, BCC and FCC structures.	3	5
3	Explain Bragg's law.	2	5
4	Derive inter-planar spacing between two successive (hkl) planes.	6	5
5	Calculate the inter-planar spacing for (3 2 1) planes in a	3	5

	simple cubic crystal whose lattice constant is 4.2 A.U.		
6	Explain crystal structure determination by Laue method.	2	5

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12. Prescribed Text Books & Reference Books

Text Book

1.A text book of Engineering Physics – Dr. M.N. Avadhanulu& Dr. P.G. Kshirsagar, S.Chand and Company, 11 Edition, 2019

2.Engineering Physics – B.K. Pandey and S. Chaturvedi, Cengage Learning,2013

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